

Student name:

**CCNA v7.0 –SEMETER 1
FINAL SKILL EXAM**

Exam ID 01

Duration: 1 hour 45 min

1. Exam objectives

- Build the network topology.
- Plan the IP addresses and apply your subnetting scheme to servers, PCs, router interfaces and switch SVI interfaces.
- Basic router and switch configuration; static routing.
- Configure DNS, HTTP and TFTP services.

2. General requirements

- Connect devices using correct interfaces and proper cables.
- Set correct hostname, assign IPv4/IPv6 addresses, subnet mask, default gateway and DNS information for every devices (*if need*).
- Every routers and switches must be set “**bkacad**” as password for console connection, vty connections and “**cisco**” as enable secret password.
- Every password should be encrypted in configuration files.
- Every router and switch’s interfaces must be set description according to Physical connection table.
- Config static route on routers to ensure connectivity between every networks.
- Enable IPv6 in VCBank network.

3. Device configuration

No	Hostname	Type	Model	Note
1	Ftel	Router	Cisco 1941	
2	VCBank	Router	Cisco 1941	
3	HDBank	Router	Cisco 1941	
4	Ftel-SW	Switch	Cisco 2960	
5	VCBank-SW1	Switch	Cisco 2960	
6	VCBank-SW2	Switch	Cisco 2960	
7	HDBank-SW	Switch	Cisco 2960	
8	Ftel-WEB	Server	Generic	enabled HTTP service
9	Ftel-DNS	Server	Generic	enabled DNS service
10	VCBank-WEB	Server	Generic	enabled HTTP service
11	VCBank-TFTP	Server	Generic	enabled TFTP service
12	VCBank-DHCP	Server	Generic	enabled DHCP service
13	HDBank-DHCP	Server	Generic	enabled DHCP service
14	PC1	PC	Generic	
15	PC2	PC	Generic	

4. Physical connection

No	First device			Last device		
	Hostname	Interface	Description	Hostname	Interface	Description
1	Ftel	S0/0/0	Connect to VCBank	VCBank	S0/0/0	Connect to Ftel
2	Ftel	S0/0/1	Connect to HDBank	HDBank	S0/0/0	Connect to Ftel
3	Ftel	G0/0	Connect to Ftel-SW	Ftel-SW	G0/1	Connect to Ftel
4	VCBank	G0/0	Connect to VCBank-SW1	VCBank-SW1	G0/1	Connect to VCBank
5	VCBank	G0/1	Connect to VCBank-SW2	VCBank-SW2	G0/1	Connect to VCBank
6	HDBank	G0/0	Connect to HDBank-SW	HDBank-SW	G0/1	Connect to HDBank
7	Ftel-SW	F0/20	Connect to Ftel-WEB	Ftel-WEB	Fa0	-
8	Ftel-SW	F0/21	Connect to Ftel-DNS	Ftel-DNS	Fa0	-
9	VCBank-SW1	F0/20	Connect to VCBank-WEB	VCBank-WEB	Fa0	-
10	VCBank-SW1	F0/21	Connect to VCBank-TFTP	VCBank-TFTP	Fa0	-
11	VCBank-SW2	F0/20	Connect to VCBank-DHCP	VCBank-DHCP	Fa0	-
12	VCBank-SW2	F0/1	Connect to PC1	PC1	Fa0	-
13	HDBank-SW	F0/20	Connect to HDBank-DHCP	HDBank-DHCP	Fa0	-
14	HDBank-SW	F0/1	Connect to PC2	PC2	Fa0	-

5. IP planning

5.1. Ftel network

No	Network description	Network	Notes
1	Ftel – VCBank	200.0.0.0/30	Ftel router use the highest usable IP in range
3	Ftel – HDBank	200.0.0.4/30	
2	Ftel LAN	222.222.222.0/24	

5.2. VCBank network

You are asked to use the 172.16.10.0/23 address space for VCBank network.

Use the following charts to create an effective addressing scheme:

No	Device	Interface	Number of hosts	Growth ratio	Note
1	VCBank	G0/0	30	20%	
2	VCBank	-	30	0%	not yet implemented
3	VCBank	G0/1	100	50%	
4	VCBank	-	100	0%	not yet implemented

No	Device	Interface	Number of hosts	Growth ratio	Note
5	VCBank	-	30	0%	not yet implemented

5.3. HDBank network

You are asked to use the 192.168.0.0/24 address space for HDBank network.

Use the following charts to create an effective addressing scheme:

No	Device	Interface	Number of hosts	Growth ratio	Note
1	HDBank	G0/0	50	20%	
2	HDBank	-	100	0%	not yet implemented

6. IP assignment rules

- Routers use the highest usable IP in range.
- Switches use the second highest usable IP in range.
- Servers are manually assigned IP address and use the next lower highest usable IP in range.
- PC1 and PC2 are configured to receive IP address from DHCP servers

7. Service configuration

7.1. DNS domain name

No	Hostname	Domain name
1	Ftel-WEB	ftel.vn
2	VCBank-WEB	vcb.com.vn
3	VCBank-TFTP	tftp.vcb.com.vn

7.2. DHCP configuration

- Pool name: VCBank_Pool and HDBank_Pool respectively.
- Declare DNS server.
- Start IP: 10th usable IP address in range.
- Maximum user: 50.

8. IPv6 Requirements

- VCBBank network is upgrading to use IPv6 with IPv4 simultaneously. This is the IPv6 address planning:

No	Device	Interface	GUA Network
1	VCBank	G0/0	2025:acad:b012:12::/64
2	VCBank	G0/1	2025:acad:b012:13::/64

- VCBank routers use the ::1 address for LAN network.
- VCBank servers are manually assigned IPv6 address.

- PC1 are automatic assigned IPv6 address use SLAAC.

9. Connectivity requirements

- Besure PCs connect successfully to every servers through it's IP addresses also domain names.
- Besure PCs access routers and switches successfully with passwords.
- Besure PC1 connect successfully to every servers in VCBank network through it's IPv6 addresses.
- Besure all routers and switches configuration are saved in NVRAM, and all VCBank's routers and switches configuration are saved in VCBank-TFTP server.